

Exercise 3

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$, two given models $\hat{f}_i : \mathcal{X} \rightarrow \mathbb{R}$ for $i = 1, 2$, and the score function

$$s(x, y) = \min_{i=1,2} |y - \hat{f}_i(x)|.$$

Derive an explicit expression for $\mathcal{C}(x)$ in terms of $\hat{f}_1(x)$, $\hat{f}_2(x)$, and the quantile $\hat{q}$ of the scores on the validation set. What is the shape of the prediction when $\hat{q}$ is small or when it is large? Briefly explain why this might be attractive if the conditional distribution of Y given $X = x$ is bimodal.

Solution By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at x , that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of Y given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////